

SAN RAFAEL, Argentina

WMO: 875090

Lat: 34.5908S

Lon: 68.4017W

Elev: 748

StdP: 92.66

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.0	-0.5	-12.4	1.4	10.9	-9.9	1.8	9.0	10.3	14.0	8.7	13.6	1.6	270	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	34.9	19.5	33.4	19.1	31.9	18.5	22.0	31.2	21.0	30.0	20.1	28.7	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	15.3	26.0	18.1	14.2	25.1	17.0	13.3	24.3	68.2	31.2	64.3	30.2	61.0	28.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.5	6.3	DB	-4.5	38.1	1.3	1.6	-5.5	39.3	-6.3	40.2	-7.0	41.2	-8.0	42.4	(4)
(5)			WB	-5.7	24.0	1.2	1.3	-6.5	25.0	-7.2	25.8	-7.8	26.5	-8.7	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	23.8	22.4	19.8	15.5	11.8	9.0	8.2	10.1	12.7	16.0	19.5	22.2	(6)	
	DBStd	6.34	2.90	3.45	3.55	3.55	3.40	3.06	3.42	3.94	3.80	3.62	3.35	3.23	(7)	
	HDD10.0	219	0	0	1	3	19	54	74	46	17	3	1	0	(8)	
	HDD18.3	1482	2	8	27	97	202	282	313	256	172	90	27	7	(9)	
	CDD10.0	2364	428	346	303	168	77	23	20	50	98	188	284	379	(10)	
	CDD18.3	584	171	121	71	12	1	0	0	1	3	17	61	128	(11)	
	CDH23.3	6912	1924	1282	751	179	22	1	3	43	94	303	790	1521	(12)	
	CDH26.7	2804	901	557	266	33	3	0	0	9	17	79	283	657	(13)	
(14) Wind	WSAvg	2.5	2.7	2.5	2.4	2.1	2.0	2.2	2.4	2.6	2.6	2.8	2.9	3.0	(14)	
(15) Precipitation	PrecAvg	336	54	47	42	24	11	10	10	15	24	31	40	45	(15)	
	PrecMax	598	204	281	178	92	53	78	51	79	113	106	160	168	(16)	
	PrecMin	107	2	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	119	45	45	37	26	13	14	13	19	25	27	32	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	36.2	33.6	29.7	25.9	22.6	22.9	27.7	28.9	31.6	34.1	36.4	(19)	
		MCWB	20.6	20.4	19.6	17.2	14.6	11.7	10.5	13.5	14.0	16.6	18.0	19.2	(20)	
	2%	DB	35.1	34.0	31.5	27.5	23.1	19.8	20.1	23.8	26.0	28.9	31.9	34.2	(21)	
		MCWB	19.9	19.6	18.9	16.4	13.9	10.3	9.8	11.9	13.1	15.6	17.5	18.7	(22)	
	5%	DB	33.2	32.1	29.7	25.3	21.0	17.8	17.9	21.0	23.7	26.7	30.0	32.2	(23)	
		MCWB	19.3	19.0	18.4	15.3	12.9	9.6	9.0	10.6	12.3	14.5	16.7	18.5	(24)	
	10%	DB	31.5	30.2	27.8	23.3	18.8	15.9	15.6	18.3	21.2	24.6	28.0	30.4	(25)	
		MCWB	18.8	18.6	17.5	14.6	11.8	8.6	7.9	9.1	11.2	13.8	15.8	17.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.2	22.2	19.3	16.2	12.8	12.1	14.5	16.1	19.1	20.8	23.0	(27)	
		MCDB	32.8	31.9	30.5	25.7	23.5	19.9	20.2	25.2	26.0	28.0	30.3	32.0	(28)	
	2%	WB	21.7	21.8	20.5	17.7	14.8	11.3	10.6	12.4	14.3	17.2	19.1	21.0	(29)	
		MCDB	31.4	30.7	28.3	24.8	21.1	18.1	18.7	22.7	23.6	25.9	28.9	30.5	(30)	
	5%	WB	20.8	20.7	19.2	16.5	13.6	10.1	9.4	11.0	13.0	15.8	17.7	19.6	(31)	
		MCDB	30.5	28.7	27.4	23.3	19.1	16.4	16.8	19.6	21.9	24.2	27.3	28.9	(32)	
	10%	WB	19.9	19.7	18.2	15.4	12.4	9.1	8.2	9.6	11.7	14.6	16.6	18.6	(33)	
		MCDB	29.0	27.4	26.2	21.7	17.4	15.2	14.6	17.3	20.0	22.5	25.9	28.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	13.5	12.8	12.3	11.6	12.6	12.9	13.5	13.6	13.6	14.6	14.5	(35)	
		MCDBR	16.8	16.5	16.1	16.0	15.2	15.7	16.5	17.7	17.9	17.6	18.1	17.4	(36)	
		MCWBR	6.4	6.2	6.6	7.6	7.9	8.8	8.9	8.8	8.5	7.9	7.9	7.1	(37)	
	5% WB	MCDBR	14.7	13.8	14.0	13.8	13.2	14.5	15.1	16.1	16.1	15.3	15.8	15.0	(38)	
		MCWBR	6.3	6.2	6.5	7.3	7.5	8.5	8.7	8.5	8.3	7.8	7.9	7.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.374	0.363	0.349	0.324	0.297	0.296	0.320	0.356	0.354	0.361	0.372	(40)	
	taud		2.361	2.400	2.426	2.434	2.467	2.546	2.517	2.438	2.338	2.374	2.370	2.373	(41)	
	Ebn at Noon		951	940	912	866	835	841	861	885	904	949	969	968	(42)	
	Edh at Noon		131	121	110	97	82	71	76	94	118	123	129	130	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.02	6.99	5.71	4.23	2.96	2.64	2.88	3.78	5.12	6.41	7.76	8.31	(44)	
	RadStd		0.44	0.38	0.41	0.41	0.34	0.22	0.20	0.28	0.34	0.63	0.45	0.42	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A	+63

Nomenclature: See separate page